QSVG-Lattice: Complete Derivatives Appendix

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Abstract

This appendix contains all explicit derivatives used in the main QSVG-Lattice framework documents. The derivatives are organized by section and numbered for cross-reference. All calculations are shown step-by-step, including first and second derivatives, critical point analysis, Hessian matrix elements, and dynamic mode equations.

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1 Derivatives of the U(1) Potential

1.1 Original Potential

$$U(\theta) = K\theta^2 \left(1 - \frac{\theta}{\theta_0}\right) \tag{1}$$

1.2 First Derivative

$$\begin{aligned} U'(\theta) &= \frac{d}{d\theta}\left[K\theta^2\left(1 - \frac{\theta}{\theta_0}\right)\right] && (2)\\ &= K\left[2\theta\left(1 - \frac{\theta}{\theta_0}\right) + \theta^2\left(-\frac{1}{\theta_0}\right)\right] && (3)\\ &= K\left[2\theta - \frac{2\theta^2}{\theta_0} - \frac{\theta^2}{\theta_0}\right] && (4)\\ &= \boxed{2K\theta - \frac{3K}{\theta_0}\theta^2} && (5) \end{aligned}$$

1.3 Second Derivative

$$U''(\theta) = \frac{d}{d\theta}\left[2K\theta - \frac{3K}{\theta_0}\theta^2\right] \tag{6}$$

$$= \boxed{2K - \frac{6K}{\theta_0}\theta} \tag{7}$$

1.4 Critical Points

$$U'(\theta) = 0 \Rightarrow 2K\theta - \frac{3K}{\theta_0}\theta^2 = 0 \tag{8}$$

$$\Rightarrow K\theta\left(2 - \frac{3\theta}{\theta_0}\right) = 0 \tag{9}$$

$$\Rightarrow \boxed{\theta = 0} \quad \text{or} \quad \boxed{\theta = \frac{2}{3}\theta_0} \tag{10}$$

1.5 Classification

$$U''(0) = 2K > 0 \Rightarrow \text{local minimum} \tag{11}$$

$$U''\left(\frac{2}{3}\theta_0\right) = 2K - \frac{6K}{\theta_0}\cdot\frac{2}{3}\theta_0 = 2K - 4K = -2K < 0 \Rightarrow \text{local maximum} \tag{12}$$

1.6 Value at Maximum

$$U_{\max} = U\left(\frac{2}{3}\theta_0\right) = K\left(\frac{4}{9}\theta_0^2\right)\left(1 - \frac{2}{3}\right) \tag{13}$$

$$= K\cdot\frac{4}{9}\theta_0^2\cdot\frac{1}{3} = \boxed{\frac{4}{27}K\theta_0^2} \tag{14}$$

2 Derivatives of the SU(2)×U(1) Potential

2.1 Original Potential

$$U_{\text{eff}}(\theta_Y, |\vec{\theta}_L|) = K_Y\theta_Y^2\left(1 - \frac{\theta_Y}{\theta_0}\right) + K_L|\vec{\theta}_L|^2\left(1 - \frac{|\vec{\theta}_L|}{\theta_0}\right) + 2\lambda\theta_Y^2|\vec{\theta}_L|^2 \tag{15}$$

Note: The factor 2λ (rather than λ) is chosen to simplify derivatives, as it yields 4λ terms in first derivatives.

2.2 Partial Derivative with Respect to θ_Y

$$\frac{\partial U}{\partial \theta_Y} = \frac{\partial}{\partial \theta_Y}\left[K_Y\theta_Y^2\left(1-\frac{\theta_Y}{\theta_0}\right)\right] + \frac{\partial}{\partial \theta_Y}\left[2\lambda\theta_Y^2|\vec{\theta}_L|^2\right] \tag{16}$$

$$= K_Y\left[2\theta_Y\left(1-\frac{\theta_Y}{\theta_0}\right) + \theta_Y^2\left(-\frac{1}{\theta_0}\right)\right] + 4\lambda\theta_Y|\vec{\theta}_L|^2 \tag{17}$$

$$= K_Y\left[2\theta_Y - \frac{2\theta_Y^2}{\theta_0} - \frac{\theta_Y^2}{\theta_0}\right] + 4\lambda\theta_Y|\vec{\theta}_L|^2 \tag{18}$$

$$= \boxed{2K_Y\theta_Y - \frac{3K_Y}{\theta_0}\theta_Y^2 + 4\lambda\theta_Y|\vec{\theta}_L|^2} \tag{19}$$

2.3 Partial Derivative with Respect to $|\vec{\theta}_L|$

$$\frac{\partial U}{\partial |\vec{\theta}_L|} = \frac{\partial}{\partial |\vec{\theta}_L|}\left[K_L|\vec{\theta}_L|^2\left(1-\frac{|\vec{\theta}_L|}{\theta_0}\right)\right] + \frac{\partial}{\partial |\vec{\theta}_L|}\left[2\lambda\theta_Y^2|\vec{\theta}_L|^2\right] \tag{20}$$

$$= K_L\left[2|\vec{\theta}_L|\left(1-\frac{|\vec{\theta}_L|}{\theta_0}\right) + |\vec{\theta}_L|^2\left(-\frac{1}{\theta_0}\right)\right] + 4\lambda\theta_Y^2|\vec{\theta}_L| \tag{21}$$

$$= K_L\left[2|\vec{\theta}_L| - \frac{2|\vec{\theta}_L|^2}{\theta_0} - \frac{|\vec{\theta}_L|^2}{\theta_0}\right] + 4\lambda\theta_Y^2|\vec{\theta}_L| \tag{22}$$

$$= \boxed{2K_L|\vec{\theta}_L| - \frac{3K_L}{\theta_0}|\vec{\theta}_L|^2 + 4\lambda\theta_Y^2|\vec{\theta}_L|} \tag{23}$$

2.4 Second Partial Derivatives

2.4.1 Second Derivative with Respect to θ_Y

$$\frac{\partial^2 U}{\partial \theta_Y^2} = \frac{\partial}{\partial \theta_Y}\left[2K_Y\theta_Y - \frac{3K_Y}{\theta_0}\theta_Y^2 + 4\lambda\theta_Y|\vec{\theta}_L|^2\right] \tag{24}$$

$$= \boxed{2K_Y - \frac{6K_Y}{\theta_0}\theta_Y + 4\lambda|\vec{\theta}_L|^2} \tag{25}$$

2.4.2 Second Derivative with Respect to $|\vec{\theta}_L|$

$$\frac{\partial^2 U}{\partial |\vec{\theta}_L|^2} = \frac{\partial}{\partial |\vec{\theta}_L|}\left[2K_L|\vec{\theta}_L| - \frac{3K_L}{\theta_0}|\vec{\theta}_L|^2 + 4\lambda\theta_Y^2|\vec{\theta}_L|\right] \tag{26}$$

$$= \boxed{2K_L - \frac{6K_L}{\theta_0}|\vec{\theta}_L| + 4\lambda\theta_Y^2} \tag{27}$$

2.4.3 Mixed Derivative

$$\frac{\partial^2 U}{\partial \theta_Y \partial |\vec{\theta}_L|} = \frac{\partial}{\partial |\vec{\theta}_L|}\left[2K_Y\theta_Y - \frac{3K_Y}{\theta_0}\theta_Y^2 + 4\lambda\theta_Y|\vec{\theta}_L|^2\right] \tag{28}$$

$$= \boxed{8\lambda\theta_Y|\vec{\theta}_L|} \tag{29}$$

By symmetry, $\frac{\partial^2 U}{\partial|\vec{\theta}_L|\partial\theta_Y} = 8\lambda\theta_Y|\vec{\theta}_L|$.

3 Derivatives with Torsion Coupling

3.1 Total Potential

$$U_{\text{total}} = U_{\text{eff}}(\theta_Y, |\vec{\theta}_L|) + \frac{\mu_T}{2}T^2 + \frac{\lambda_T}{2}|\nabla T|^2 + \gamma_Y\theta_Y T + \gamma_L|\vec{\theta}_L|^2 T \tag{30}$$

3.2 Derivative with Respect to θ_Y (with coupling)

$$\frac{\partial U_{\text{total}}}{\partial\theta_Y} = \frac{\partial U_{\text{eff}}}{\partial\theta_Y} + \gamma_Y T = \boxed{2K_Y\theta_Y - \frac{3K_Y}{\theta_0}\theta_Y^2 + 4\lambda\theta_Y|\vec{\theta}_L|^2 + \gamma_Y T} \tag{31}$$

3.3 Derivative with Respect to $|\vec{\theta}_L|$ (with coupling)

$$\frac{\partial U_{\text{total}}}{\partial|\vec{\theta}_L|} = \frac{\partial U_{\text{eff}}}{\partial|\vec{\theta}_L|} + 2\gamma_L|\vec{\theta}_L|T = \boxed{2K_L|\vec{\theta}_L| - \frac{3K_L}{\theta_0}|\vec{\theta}_L|^2 + 4\lambda\theta_Y^2|\vec{\theta}_L| + 2\gamma_L|\vec{\theta}_L|T} \tag{32}$$

3.4 Derivative with Respect to T

3.4.1 Ordinary Partial Derivative

$$\frac{\partial U_{\text{total}}}{\partial T} = \mu_T T + \gamma_Y\theta_Y + \gamma_L|\vec{\theta}_L|^2 \tag{33}$$

3.4.2 Functional Derivative with Gradient Term

For the functional derivative including the gradient term:

$$\frac{\delta}{\delta T}\int \frac{\lambda_T}{2}|\nabla T|^2 d^3x = -\lambda_T\nabla^2 T \tag{34}$$

Therefore, the total functional derivative is:

$$\boxed{\frac{\delta U_{\text{total}}}{\delta T} = \mu_T T - \lambda_T\nabla^2 T + \gamma_Y\theta_Y + \gamma_L|\vec{\theta}_L|^2} \tag{35}$$

4 Hessian Matrix (Second Derivatives with Torsion)

4.1 Definition

$H_{ij} = \frac{\partial^2 U_{\text{total}}}{\partial\Phi_i\partial\Phi_j}$ where $\Phi = (\theta_Y, |\vec{\theta}_L|, T)$.

4.2 Element H_{11}

$$H_{11} = \frac{\partial^2 U_{\text{total}}}{\partial\theta_Y^2} = \frac{\partial}{\partial\theta_Y}\left[2K_Y\theta_Y - \frac{3K_Y}{\theta_0}\theta_Y^2 + 4\lambda\theta_Y|\vec{\theta}_L|^2 + \gamma_Y T\right] \tag{36}$$

$$= \boxed{2K_Y - \frac{6K_Y}{\theta_0}\theta_Y + 4\lambda|\vec{\theta}_L|^2} \tag{37}$$

4.3 Element H_{22}

$$H_{22} = \frac{\partial^2 U_{\text{total}}}{\partial |\vec{\theta}_L|^2} = \frac{\partial}{\partial |\vec{\theta}_L|} \left[2K_L |\vec{\theta}_L| - \frac{3K_L}{\theta_0} |\vec{\theta}_L|^2 + 4\lambda \theta_Y^2 |\vec{\theta}_L| + 2\gamma_L |\vec{\theta}_L| T \right] \tag{38}$$

$$= \boxed{2K_L - \frac{6K_L}{\theta_0} |\vec{\theta}_L| + 4\lambda \theta_Y^2 + 2\gamma_L T} \tag{39}$$

4.4 Element H_{33}

For constant modes (ignoring spatial derivatives):

$$H_{33} = \frac{\partial^2 U_{\text{total}}}{\partial T^2} = \boxed{\mu_T} \tag{40}$$

For spatial modes, the Hessian in Fourier space becomes $H_{33} = \mu_T + \lambda_T k^2$.

4.5 Element $H_{12} = H_{21}$

$$H_{12} = \frac{\partial^2 U_{\text{total}}}{\partial \theta_Y \partial |\vec{\theta}_L|} = \frac{\partial}{\partial |\vec{\theta}_L|} \left[2K_Y \theta_Y - \frac{3K_Y}{\theta_0} \theta_Y^2 + 4\lambda \theta_Y |\vec{\theta}_L|^2 + \gamma_Y T \right] \tag{41}$$

$$= \boxed{8\lambda \theta_Y |\vec{\theta}_L|} \tag{42}$$

4.6 Element $H_{13} = H_{31}$

$$H_{13} = \frac{\partial^2 U_{\text{total}}}{\partial \theta_Y \partial T} = \frac{\partial}{\partial T} \left[2K_Y \theta_Y - \frac{3K_Y}{\theta_0} \theta_Y^2 + 4\lambda \theta_Y |\vec{\theta}_L|^2 + \gamma_Y T \right] \tag{43}$$

$$= \boxed{\gamma_Y} \tag{44}$$

4.7 Element $H_{23} = H_{32}$

$$H_{23} = \frac{\partial^2 U_{\text{total}}}{\partial |\vec{\theta}_L| \partial T} = \frac{\partial}{\partial T} \left[2K_L |\vec{\theta}_L| - \frac{3K_L}{\theta_0} |\vec{\theta}_L|^2 + 4\lambda \theta_Y^2 |\vec{\theta}_L| + 2\gamma_L |\vec{\theta}_L| T \right] \tag{45}$$

$$= \boxed{2\gamma_L |\vec{\theta}_L|} \tag{46}$$

4.8 Complete Hessian Matrix

$$\boxed{H = \begin{pmatrix} 2K_Y - \frac{6K_Y}{\theta_0} \theta_Y + 4\lambda |\vec{\theta}_L|^2 & 8\lambda \theta_Y |\vec{\theta}_L| & \gamma_Y \\ 8\lambda \theta_Y |\vec{\theta}_L| & 2K_L - \frac{6K_L}{\theta_0} |\vec{\theta}_L| + 4\lambda \theta_Y^2 + 2\gamma_L T & 2\gamma_L |\vec{\theta}_L| \\ \gamma_Y & 2\gamma_L |\vec{\theta}_L| & \mu_T \end{pmatrix}} \tag{47}$$

5 Derivatives for Dynamic Modes with Photon Viscosity

5.1 Lagrangian with Kinetic Terms

$$\mathcal{L} = \frac{\eta_\gamma}{2} (\partial_t \theta_Y)^2 + \frac{\eta_\gamma}{2} |\partial_t \vec{\theta}_L|^2 - U_{\text{total}}(\theta_Y, |\vec{\theta}_L|, T) \tag{48}$$

5.2 Euler-Lagrange Equations

5.2.1 For θ_Y

$$\frac{\delta \mathcal{L}}{\delta \theta_Y} = -\eta_\gamma \partial_{tt}\theta_Y - \frac{\partial U_{\text{total}}}{\partial \theta_Y} = 0 \tag{49}$$

$$\Rightarrow \boxed{\eta_\gamma \partial_{tt}\theta_Y + \frac{\partial U_{\text{total}}}{\partial \theta_Y} = 0} \tag{50}$$

5.2.2 For $|\vec{\theta}_L|$

$$\frac{\delta \mathcal{L}}{\delta |\vec{\theta}_L|} = -\eta_\gamma \partial_{tt}|\vec{\theta}_L| - \frac{\partial U_{\text{total}}}{\partial |\vec{\theta}_L|} = 0 \tag{51}$$

$$\Rightarrow \boxed{\eta_\gamma \partial_{tt}|\vec{\theta}_L| + \frac{\partial U_{\text{total}}}{\partial |\vec{\theta}_L|} = 0} \tag{52}$$

5.2.3 For T

$$\frac{\delta \mathcal{L}}{\delta T} = -\frac{\delta U_{\text{total}}}{\delta T} = 0 \tag{53}$$

$$\Rightarrow \boxed{\frac{\delta U_{\text{total}}}{\delta T} = 0} \tag{54}$$

5.3 Linearized Equations

Expand around a critical point $\Phi_0 = (\theta_Y^0, |\vec{\theta}_L|^0, T^0)$:

$$\theta_Y = \theta_Y^0 + \delta\theta_Y e^{i(\omega t - \vec{k}\cdot\vec{r})} \tag{55}$$

$$|\vec{\theta}_L| = |\vec{\theta}_L|^0 + \delta|\vec{\theta}_L| e^{i(\omega t - \vec{k}\cdot\vec{r})} \tag{56}$$

$$T = T^0 + \delta T e^{i(\omega t - \vec{k}\cdot\vec{r})} \tag{57}$$

5.4 Linearized Field Equations

$$(-\eta_\gamma \omega^2)\delta\theta_Y + H_{11}\delta\theta_Y + H_{12}\delta|\vec{\theta}_L| + H_{13}\delta T = 0 \tag{58}$$

$$(-\eta_\gamma \omega^2)\delta|\vec{\theta}_L| + H_{21}\delta\theta_Y + H_{22}\delta|\vec{\theta}_L| + H_{23}\delta T = 0 \tag{59}$$

$$(-\lambda_T k^2 - \mu_T)\delta T + H_{31}\delta\theta_Y + H_{32}\delta|\vec{\theta}_L| = 0 \tag{60}$$

5.5 Characteristic Equation

$$\det \begin{pmatrix} H_{11} - \eta_\gamma \omega^2 & H_{12} & H_{13} \\ H_{21} & H_{22} - \eta_\gamma \omega^2 & H_{23} \\ H_{31} & H_{32} & \mu_T + \lambda_T k^2 \end{pmatrix} = 0 \tag{61}$$

This determinant yields the dispersion relations $\omega(k)$ for the coupled system.

6 Derivatives for Mass Corrections

6.1 Torsion Mass Correction (W boson)

$$\Delta M_W^{\text{torsion}} = \left.\frac{\partial^2 U_{\text{coupled}}}{\partial|\vec{\theta}_L|^2}\right|_{\min} L^3 \tag{62}$$

$$\frac{\partial^2 U_{\text{coupled}}}{\partial|\vec{\theta}_L|^2} = \frac{\partial}{\partial|\vec{\theta}_L|}\left[2\gamma_L|\vec{\theta}_L|T\right] = 2\gamma_L T \tag{63}$$

$$\Rightarrow \Delta M_W^{\text{torsion}} = \boxed{2\gamma_L T^0 L^3} \tag{64}$$

6.2 Torsion Mass Correction (Z boson)

$$\Delta M_Z^{\text{torsion}} = \left.\frac{\partial^2 U_{\text{coupled}}}{\partial\theta_Y^2}\right|_{\min} L^3 \tag{65}$$

$$\frac{\partial^2 U_{\text{coupled}}}{\partial\theta_Y^2} = \frac{\partial}{\partial\theta_Y}\left[\gamma_Y T\right] = 0 \tag{66}$$

$$\Rightarrow \Delta M_Z^{\text{torsion}} = \boxed{0 \text{ (direct)}} \tag{67}$$

However, the Z boson receives indirect corrections through the shifted vacuum expectation values θ_Y^0 and $|\vec{\theta}_L|^0$.

6.3 Dynamic Mass Correction

From the linearized equations, the dynamic correction to the mass squared is:

$$\Delta M_{\text{dynamic}}^2 = \boxed{\eta_\gamma \omega_0^2} \tag{68}$$

where ω_0 is the frequency of the mode corresponding to the boson field, obtained from the characteristic equation.

7 Summary Table of All Derivatives

8 Conclusion

This appendix provides all explicit derivatives required for the QSVG-Lattice framework, including:

- U(1) vortex potential derivatives with critical point analysis
- SU(2)×U(1) potential partial derivatives
- Torsion-coupled potential derivatives
- Complete 3×3 Hessian matrix
- Dynamic mode equations with photon viscosity
- Mass correction formulas

All calculations are shown step-by-step, making this appendix a complete reference for numerical implementation and further theoretical development.

Derivative	**Expression**
$dU_{U(1)}/d\theta$	$2K\theta - \frac{3K}{\theta_0}\theta^2$
$d^2U_{U(1)}/d\theta^2$	$2K - \frac{6K}{\theta_0}\theta$
$\partial U_{SU(2)\times U(1)}/\partial\theta_Y$	$2K_Y\theta_Y - \frac{3K_Y}{\theta_0}\theta_Y^2 + 4\lambda\theta_Y\lvert\vec{\theta}_L\rvert^2$
$\partial U_{SU(2)\times U(1)}/\partial\lvert\vec{\theta}_L\rvert$	$2K_L\lvert\vec{\theta}_L\rvert - \frac{3K_L}{\theta_0}\lvert\vec{\theta}_L\rvert^2 + 4\lambda\theta_Y^2\lvert\vec{\theta}_L\rvert$
$\partial^2U/\partial\theta_Y^2$	$2K_Y - \frac{6K_Y}{\theta_0}\theta_Y + 4\lambda\lvert\vec{\theta}_L\rvert^2$
$\partial^2U/\partial\lvert\vec{\theta}_L\rvert^2$	$2K_L - \frac{6K_L}{\theta_0}\lvert\vec{\theta}_L\rvert + 4\lambda\theta_Y^2$
$\partial^2U/\partial\theta_Y\partial\lvert\vec{\theta}_L\rvert$	$8\lambda\theta_Y\lvert\vec{\theta}_L\rvert$
$\partial U_{\text{total}}/\partial\theta_Y$ (with torsion)	$2K_Y\theta_Y - \frac{3K_Y}{\theta_0}\theta_Y^2 + 4\lambda\theta_Y\lvert\vec{\theta}_L\rvert^2 + \gamma_Y T$
$\partial U_{\text{total}}/\partial\lvert\vec{\theta}_L\rvert$ (with torsion)	$2K_L\lvert\vec{\theta}_L\rvert - \frac{3K_L}{\theta_0}\lvert\vec{\theta}_L\rvert^2 + 4\lambda\theta_Y^2\lvert\vec{\theta}_L\rvert + 2\gamma_L\lvert\vec{\theta}_L\rvert T$
$\delta U_{\text{total}}/\delta T$	$\mu_T T - \lambda_T\nabla^2 T + \gamma_Y\theta_Y + \gamma_L\lvert\vec{\theta}_L\rvert^2$
H_{11}	$2K_Y - \frac{6K_Y}{\theta_0}\theta_Y + 4\lambda\lvert\vec{\theta}_L\rvert^2$
H_{22}	$2K_L - \frac{6K_L}{\theta_0}\lvert\vec{\theta}_L\rvert + 4\lambda\theta_Y^2 + 2\gamma_L T$
H_{33}	μ_T (constant mode)
H_{12}	$8\lambda\theta_Y\lvert\vec{\theta}_L\rvert$
H_{13}	γ_Y
H_{23}	$2\gamma_L\lvert\vec{\theta}_L\rvert$

Table 1: Summary of all explicit derivatives used in the QSVG-Lattice framework.